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Photoredox-Catalyzed Hydrosulfonylation of Arylallenes

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Supporting Information

ABSTRACT: (Het)Arylallenes undergo hydrosulfonylation under photoredox-catalyzed conditions. The reaction gives vinyl sulfones in a regio- and diastereoselective manner, employing sodium sulfinates as the sulfonyl source and eosin Y as the photocatalyst. Indol-1-yl, pyrrol-1-yl, phenyl, and naphtylallenes might be used. Aliphatic allenes are incompatible with the reaction conditions.



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INTRODUCTION

The emerging field of photoredox catalysis has become an indispensable tool for organic synthesis, enabling novel and intrinsic ways of bond creation.[1] Photoredox-catalyzed reactions proceed mainly through single-electron transfer processes, giving rise to odd electron chemistry. The use of a renewable resource of visible light to facilitate chemical reactions allows the transformations under mild conditions to be performed and the use of stoichiometric amounts of tin to be evaded, usually employed in free radical chemistry. Beyond Ru, Ir, and Cu complexes as photocatalysts, the employment of organic dyes seems a practical and green alternative.[2] For instance, the xanthene dye eosin Y has a much lower cost and toxicity, with comparable and sometimes superior effectiveness.[3]

Vinyl sulfones are sought for compounds for medicine and biology, as they display extensive biological activity[4] and are excellent substrates for the labeling of biomolecules.[5] For instance, K11777 is a renowned cysteine protease inhibitor[6] and sulfone **A** is an Nrf activator with potency to become a therapeutic for Parkinson's disease.[7] The combination of the sulfonyl group with an indole moiety gives rise to a class of anti-HIV-1 reverse transcriptase inhibitors[8] (Figure 1).

Vinyl sulfones are also widely employed as useful synthetic intermediates.[9] The synthesis of sulfones is a needful task, with numerous methods developed.[10] In general, the preparation of vinyl sulfones relies on sulfonylation of alkenes through oxidative[11] or cross-coupling reactions,[12] decarboxylative couplings with alkenes[13] or alkynes,[14] tosylhydrazone decompositions,[15] SO_2 fixation approaches,[16] or hydrosulfonylation of alkynes.[17] The metal-free synthesis of sulfones is also realizable from sulfonate esters.[18] The common drawbacks of these methods are the employment of transition-metal catalysts and/or stoichiometric oxidants and the use of sacrificial

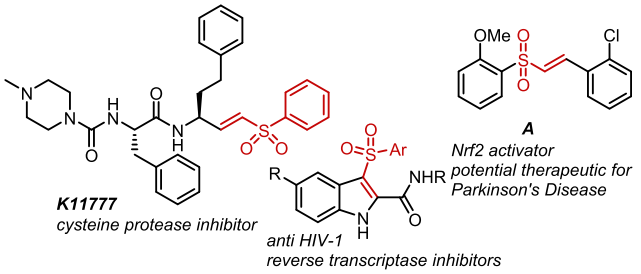


Figure 1. Representative biologically active sulfones.

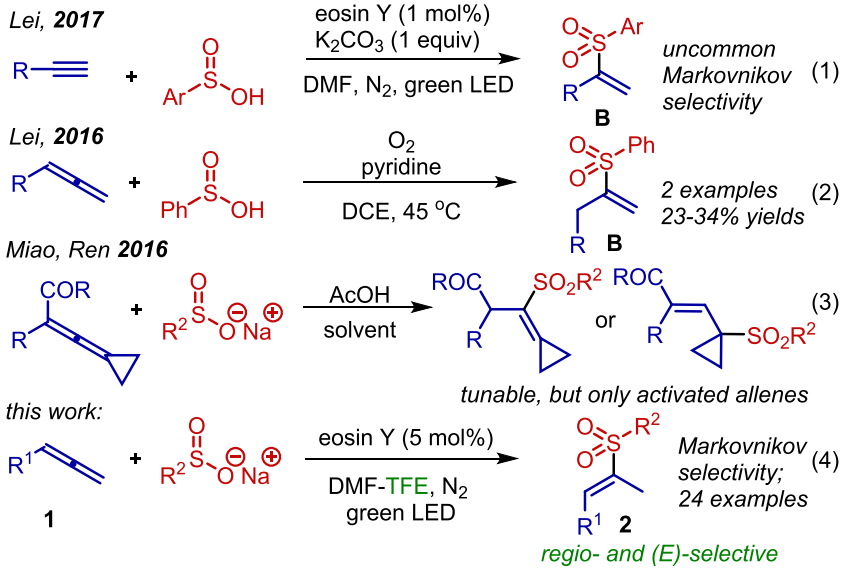
groups. Recently, photoredox-catalyzed hydrosulfonylation of alkynes has been reported by Lei and co-workers.[19] Uncommon for radical processes, Markovnikov selectivity has been observed, producing branched vinyl sulfones **B** (Scheme 1, eq 1). Allenes are a readily available class of compounds and are widely used in the synthesis of natural products, pharmaceuticals, and materials.[20] Therefore, the development of novel methods for selective allene functionalization is an appealing task for organic chemists. Free radical transformations of allenes are developing rapidly,[21] and photoredox catalysis is a sustainable option for this. However, free radical hydrosulfonylation of allenes has been realized in few examples (Scheme 1, eq 2).[22] Anionic hydrosulfonylation has been developed by Miao, Ren, and co-workers but is limited to unique highly activated allenes (Scheme 1, eq 3).[23] Herein, we report a photoredox-catalyzed regio- and diastereoselective hydrosulfonylation of inactivated allenes **1** toward Markovnikov-type vinyl sulfones **2**, which we believe

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Scheme 1. Recent Examples of Vinyl Sulfone Syntheses through Hydrosulfonylation

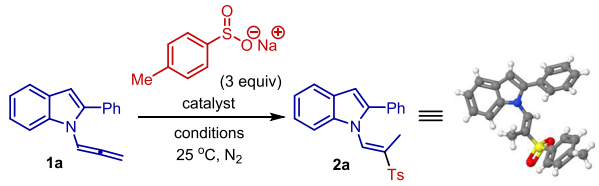


excellently complements the existing methods (Scheme 1, eq 4).

RESULTS AND DISCUSSION

Our recent analysis of the impact of photocatalyzed reactions on the field of indole chemistry uncovered the lack of methods where the indole moiety is not modified.[24] We became interested in the outcome of the reaction of 2-phenyl-*N*-(1,2-propadien-1-yl)indole **1a** under photocatalyzed conditions. We first examined the reaction of 2-phenyl-*N*-(1,2-propadien-1-yl)indole **1**a with sodium *p*-tolylsulfinate in the presence of eosin Y in a DMF−water mixture under visible light irradiation (Table 1, entry 1). To our delight, we isolated vinyl sulfone **2a** as the major product in 40% yield. The use of a larger amount of the catalyst and a smaller amount of water (5 equiv) yielded vinyl sulfone **2a** in 62% yield (Table 1, entry 2). The reaction did not start without the addition of water (Table 1, entry 3) or without irradiation with visible light (Table 1, entry 4). The use of Rose Bengal (Table 1, entry 5) or a $Ru(bpy)_3Cl_2$ complex (Table 1, entry 6) as a photocatalyst sped up the formation of **2a**, but no improvement of the yield was observed. Basic medium was produced through the continuous formation of hydroxide anions by considering water to be the proton source. We succeeded in identifying 2-phenylindole to be the major byproduct of the reaction. It might arise from the basic hydrolysis of the product **2a**. To suppress its formation, we explored various hydrogen sources. In ethanol, the reaction was sluggish, and the target molecule was isolated in a mere 35% yield (Table 1, entry 7). The addition of chloroform (Table 1, entry 8) instead of water gave product **2** in 46% yield. Employment of less nucleophilic alcohols like TFE and HFIP raised the yield to 61 and 56%, respectively (Table 1, entries 9 and 10). The addition of base, together with the employment of TFE, was identified as the condition of choice due to the good yield and the ease of isolation (Table 1, entry 11). Carrying out the reaction in TFE solely led to a decreased yield of 38% (Table 1, entry 12). In the absence of the catalyst, the reaction did not start either with or without light (Table 1, entries 13 and 14). The structure of compound **2a** was unambiguously determined by a single-crystal X-ray diffraction study (for details, see CCDC #1922457).

Table 1. Optimization of Hydrosulfonylation of 2-Phenyl-*N*-(1,2-propadien-1-yl)indole[a]



entry	catalyst (mol %)	conditions	yield **2a**, %[b]
1	eosin Y (3)	DMF−H_2O (3:1), 5 h, green LED	40
2	eosin Y (5)	DMF−H_2O (5 equiv), 14 h, green LED	62
3	eosin Y (5)	DMF, 20 h, green LED	0
4[c]	eosin Y (5)	DMF−H_2O (5 equiv), 14 h, no light	0
5	Rose bengal (10)	DMF−H_2O (5 equiv), 1 h, green LED	60
6[d]	$Ru(bpy)_3Cl_2·6H_2O$ (3)	DMF−H_2O (5 equiv), 10 h, blue LED	52
7	eosin Y (5)	EtOH, 14 h, green LED	35
8	eosin Y (5)	DMF−$CHCl_3$ (3:1), 6 h, green LED	46
9	eosin Y (5)	DMF−TFE (1:0.3), 4 h, green LED	61
10	eosin Y (5)	DMF−HFIP (2:1), 5 h, green LED	56
11[e]	eosin Y (5)	DMF−TFE (1:0.3), K_2CO_3, 4 h, green LED	63
12	eosin Y (5)	TFE, K_2CO_3, 4 h, green LED	38
13	no catalyst	DMF−TFE (1:0.3), 4 h, green LED	0
14	no catalyst	DMF−TFE (1:0.3), 4 h, no light	0

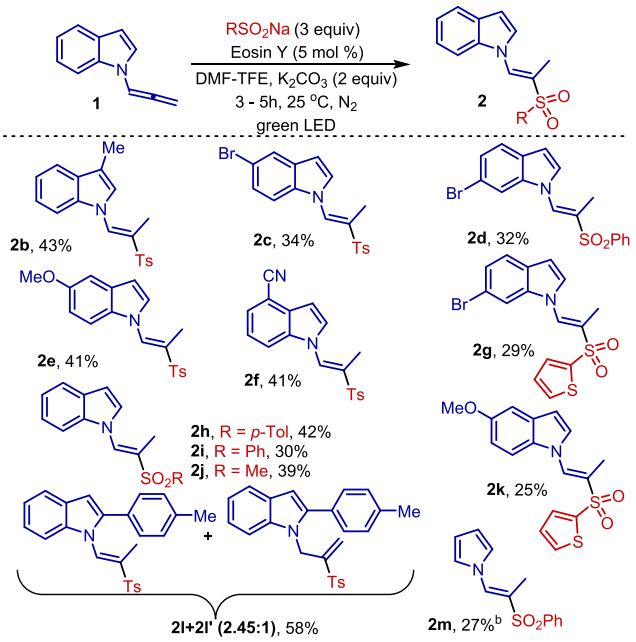
[a]General conditions: allene **1a** (0.581 mmol), sodium sulfinate (1.742 mmol), eosin Y (0.029 mmol), and the solvent (0.092 M) under a nitrogen atmosphere were irradiated with one 30 W LED or two 15 W 530 nm LEDs for 1−14 h. [b]Yields after column chromatography. [c]No light. [d]One 30 W 455 nm LED was used. [e]K_2CO_3 (2 equiv) was added to the reaction mixture.

With the optimized conditions in hand, we determined the scope of the process. We were pleased to find that indoles

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bearing a methyl, a bromo, a methoxy, or a cyano group gave products **2b–f** in 32–43% yields (Scheme 2). Unsubstituted

Scheme 2. Photoredox-Catalyzed Hydrosulfonylation of *N*-(1,2-Propadien-1-yl)indoles[a,b]

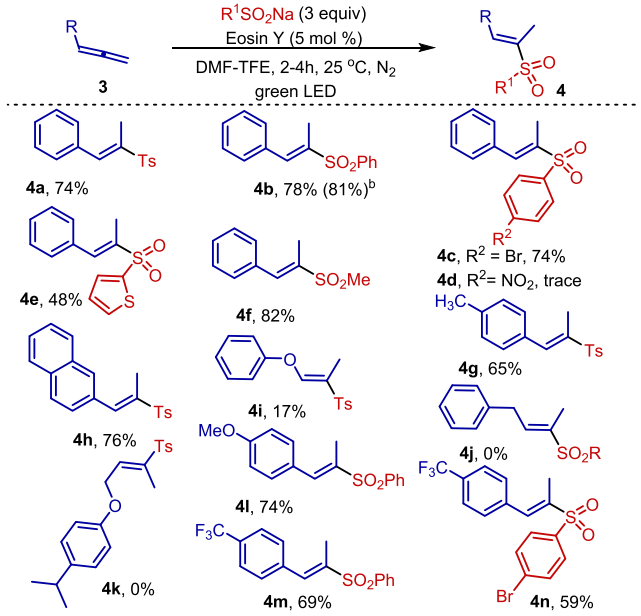


[a]General conditions: allene **1** (0.581 mmol), sodium sulfinate (1.742 mmol), eosin Y (0.029 mmol), K_2CO_3 (1.161 mmol), 4.850 mL of DMF, and 1.455 mL of TFE (1:0.3) under a nitrogen atmosphere were irradiated with one 30 W LED or two 15 W 530 nm LEDs for 3–5 h (TLC tracking). The products were isolated by column chromatography. [b]The reaction was performed on 4.76 mmol scale of allene.

indolyl allene could be hydrosulfonylated to give product **2h** with 42% yield. The employment of various sulfinates was also possible. Thus, products **2g, i, k** with thienyl or phenyl substituents in the sulfonyl moiety were synthesized. It is worth noting that aliphatic sulfinate $MeSO_2Na$ was successfully used to produce the corresponding vinyl sulfone **2j** with 39% yield. Indole with a *p*-tolyl group at C(2) underwent the discovered transformation, producing a mixture of isomers **2l** and **2l′** with moderate 58% yield. The reaction of *N*-(allenyl)pyrrole was also successful, and the corresponding product **2n** was isolated with 27% yield. As far as photoredox-catalyzed C(3)-sulfonylation of indoles was previously reported,[25] it is of special interest that in our reactions no C(3)-sulfonylated indoles were observed.

To better understand the scope of the discovered process, we investigated the reaction of 1-phenylpropadiene-1,2 (**3a**). Fortunately, employing our optimized conditions, vinyl sulfone **4a** was provided with a good yield of 74% (Scheme 3). Phenylallene **3a** reacted smoothly with phenyl-, 4-bromophenyl-, thien-2-yl-, and methylsulfinates affording sulfones **4b, c, e, f** with 48–82% yields. The interaction of **3a** with sodium *p*-nitrophenylsulfinate gave the target product **4d** in trace amounts. The use of sodium trifluoromethanesulfinate (Langlois reagent), having higher oxidative potential than aromatic or aliphatic sulfinates, led to zero conversion of the starting allene **3a**. 4-Methylphenylallene and naphtylallene could be used successfully to give products **4g** and **4h**,

Scheme 3. Photoredox-Catalyzed Hydrosulfonylation of Arylallenes[a,b]



[a]General conditions: allene **3** (0.603 mmol), sodium sulfinate (1.810 mmol), eosin Y (0.0302 mmol), 5.0 mL of DMF, and 1.5 mL of TFE (1:0.3) under a nitrogen atmosphere were irradiated with one 30 W LED or two 15 W 530 nm LEDs for 2–4 h (TLC tracking). The products were isolated by column chromatography. [b]The yield of the reaction, performed on 2 mmol scale of allene.

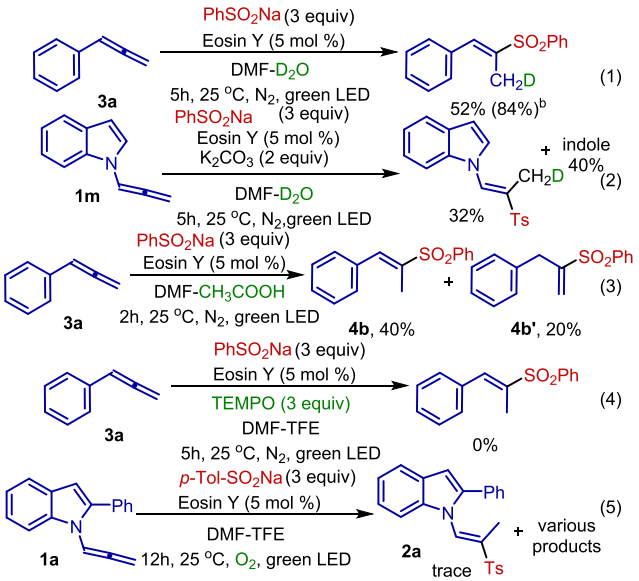
respectively. The yield of phenoxy-substituted sulfone **4i** was substantially lower. Finally, we found the reaction scope to be limited to conjugated allenes only. Thus, the reaction of benzylallene resulted in a complex inseparable mixture.

To clarify the process pathway, we performed some control experiments. First, the reaction of phenylallene **3a** in the presence of D_2O was carried out (Scheme 4, eq 1). The resulting vinyl sulfone showed incorporated deuterium exclusively in the methyl group. The analogous outcome was observed for the reaction of allenyl indole **1m** in D_2O (Scheme 4, eq 2). When the water was taken in a stoichiometric amount, but not as a solvent, the deuterium could also be observed in the α-position (see the Supporting Information). Interestingly, when the reaction was carried out in the presence of acetic acid, the reaction was faster and a mixture of regioisomeric vinyl sulfones **4b** and **4b′** (2:1) was formed (Scheme 4, eq 3). The complete quenching of the reaction in the presence of TEMPO proved the radical nature of the process (Scheme 4, eq 4). Moreover, the reaction was significantly inhibited when carried out under an oxygen atmosphere and the target product **2a** was formed in trace amounts as an inseparable mixture with numerous byproducts (Scheme 4, eq 5).

Additional mechanistic investigations to better understand the reaction were performed. The photoexcited state of EY was found to be quenched by sulfinate. On the opposite, no fluorescence quenching of EY was observed with phenylallene (for more details, see the Supporting Information). The quantum yield of the reaction was determined to be 0.88 (for more details, see the Supporting Information). High photo-effectiveness was in accordance with fast reaction times, but the chain process was unlikely. According to the literature,[26] mechanistic investigations, and the control experiments, we

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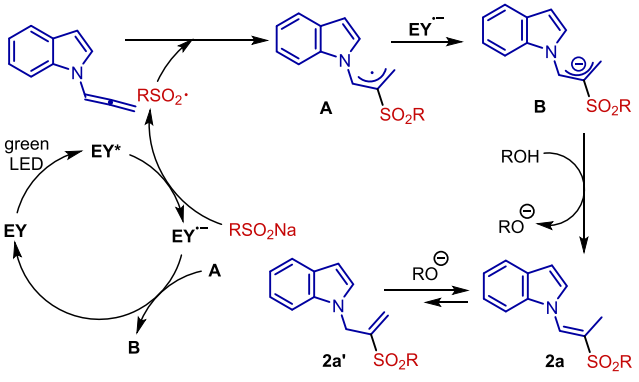
Scheme 4. Control Experiments[a,b]



[a]General conditions: allene **3** (0.431 mmol), sodium sulfinate (1.293 mmol), eosin Y (0.022 mmol), 3.6 mL of DMF, and 1.6 mL of TFE under a nitrogen atmosphere were irradiated with one 30 W LED or two 12 W 530 nm LEDs for 2−12 h. The products were isolated by column chromatography. [b]The yield after recovering the starting allene.

proposed the following mechanism for the allene hydrosulfonylation. The photoexcited state of EY was reductively quenched by sodium sulfinate, and a sulfonyl radical was produced. The addition of this sulfonyl radical to the allene system resulted in the formation of allyl radical **A**. It might be reduced to anion **B** by the EY anion radical. Further abstraction of a proton from water or alcohol delivered the final compound (Scheme 5). The selective formation of **2a** in most cases might be explained by thermodynamically driven base-catalyzed isomerization of **2a′**.

Scheme 5. Proposed Reaction Pathway



In conclusion, we have elaborated on a novel photocatalyzed protocol for the hydrosulfonylation of allenes. The reaction employs readily available sodium sulfinates as the sulfonyl source and eosin Y as the photocatalyst. We successfully applied this process for the preparation of indolyl- and aryl-substituted vinyl sulfones. The reaction tolerates various sulfinates and conjugated allenes, while the use of aliphatic allenes has been found to be a limitation of the method.

EXPERIMENTAL SECTION

General Information. All solvents were dried and distilled according to the standard procedures before use. Reagents and catalysts were purchased from the commercial sources and used without additional purification. Sodium sulfinates were synthesized according to the literature procedure.[27] ^{1}H and ^{13}C NMR spectra were recorded on 400 and 600 MHz spectrometers and referenced to the residual signals of the solvent (for ^{1}H and ^{13}C). Chemical shifts are reported in parts per million (δ/ppm). Coupling constants are reported in hertz (J/Hz). The peak patterns are indicated as follows: s, singlet; d, doublet; t, triplet; q, quadruplet; m, multiplet; dd, doublet of doublets; and br s, broad singlet. Infrared spectra were recorded on an FT-IR instrument. The wavelengths are reported in reciprocal centimeters (λ_{max}/cm^{-1}). High-resolution mass spectra (HRMS) were recorded by electrospray ionization using an LTQ Orbitrap Elite ThermoScientific mass spectrometer. The reaction progress was monitored by TLC, and the spots were visualized under UV light (254 or 365 nm). Column chromatography was performed using silica gel (230−400 mesh) and mixtures in different proportions of ethyl acetate with hexane as the mobile phase. Melting points were determined on an SMP-10 apparatus.

Synthesis of 2-Substituted *N*-Allenylindoles 1a,b. 2-Substituted indolylallenes were obtained from corresponding propargylindoles with KOH and TBAB in toluene. The alkylation procedure, based on the use of indoles, propargyl bromide, and the KOH/DMSO system (see below for **1c−i** compounds), is not selective and leads to the mixtures of the corresponding propargyl and allenylindoles.

*2-Phenyl-1-propadienyl-1H-indole (**1a**).* To a solution of 2-phenyl-*N*-propargyl indole (0.500 g, 2.2 mmol) in toluene (27 mL), 0.071 g of TBAB (0.1 equiv, 0.22 mmol) and 0.121 g of KOH (1 equiv, 2.2 mmol) were added. The mixture was stirred at rt for 24 h. Allene was used in the further reactions after filtration and evaporation of the solvent, considering the reaction yield to be quantitative. The crude allene was then chromatographed through a thin layer of silica neutralized with Et_3N, eluent EtOAc−hexane 1−10, to obtain the analytical sample. Yellow oil. R_f = 0.4 (EtOAc−hexane 1−20). ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.55 (d, J = 6.5 Hz, 2H), 6.73 (s, 1H), 7.15 (t, J = 7.5 Hz, 1H), 7.22 (t, J = 7.8 Hz, 1H), 7.25 (t, J = 6.5 Hz, 1H), 7.43−7.47 (m, 1H), 7.52 (t, J = 7.6 Hz, 2H), 7.60−7.62 (m, 3H), 7.78 ppm (d, J = 8.3 Hz, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 86.4, 95.9, 103.7, 111.3, 120.5, 120.8, 122.2, 128.2, 128.3, 128.6 (2C), 129.1 (2C), 131.6, 137.0, 139.7, 204.4. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{17}H_{14}N$: 232.1120; found: 232.1121. IR (KBr): 3054−3031, 2954−2852, 1739−1606, 1462, 1378, 745, 698 cm^{-1}.

*2-(4-Methylphenyl)-1-propadienyl-1H-indole (**1b**).* To a solution of 2-(4-methylphenyl)-*N*-propargyl indole (0.300 g, 1.2 mmol) in toluene (16 mL), 0.039 g of TBAB (0.1 equiv, 0.12 mmol) and 0.069 g of KOH (1 equiv, 1.2 mmol) were added. The mixture was stirred at rt for 24 h. Allene was used in the further reactions after filtration and evaporation of the solvent, considering the reaction yield to be quantitative. The crude allene was then chromatographed through a thin layer of silica neutralized with Et_3N, eluent EtOAc−hexane 1−10, to obtain the analytical sample. Yellow liquid. R_f = 0.55 (EtOAc−hexane 1−10). ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.45 (s, 3H), 5.52 (d, J = 6.5 Hz, 2H), 6.63 (s, 1H), 7.08 (t, J = 6.5 Hz, 1H), 7.20 (m, 1H), 7.25 (m, 1H), 7.30 (d, J = 7.8 Hz, 2H), 7.49 (d, J = 7.8 Hz, 2H), 7.64 (d, J = 7.8 Hz, 1H), 7.78 ppm (d, J = 8.1 Hz, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 21.3, 86.1, 96.8, 103.6, 111.7, 120.5, 120.9, 122.1, 129.0, 129.2 (2C), 129.3, 129.6 (2C), 137.2, 138.1, 140.4, 204.7. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{18}H_{16}N$: 246.1277; found: 246.1269. IR (KBr): 3094−3028, 2972−2920, 2859, 1706, 1502, 1458, 1358−1322, 823, 787, 748 cm^{-1}.

General Procedure for the Synthesis of 1-Propadienyl-1*H*-indoles or Pyrroles 1c−i. To a solution of indole (1.282 mmol, 1 equiv) in 2.5 mL of DMSO, 0.360 g of KOH (5 equiv) was added, and the mixture was stirred for 45 min at rt. Then, 0.158 mL of propargyl bromide (80% solution in toluene, 1.1 equiv) was added slowly (28−30 °C, exothermic reaction), and the resulting brown reaction mixture was stirred at rt for 45−60 min (TLC, EtOAc−

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hexane 1−10). Upon reaction completion, the reaction mixture was diluted with water and extracted with Et_2O or EtOAc, and the organic layer was then washed with water until the neutral reaction of the water layer. The organic phase was then dried with Na_2SO_4 and evaporated under reduced pressure. The residue was then used in the reactions without further purification. The allenes were rapidly chromatographed through a thin layer of silica gel, preliminarily neutralized with Et_3N, eluent EtOAc−hexane in proportions 1−20 or 1−10, to obtain analytically pure samples.

*1-Propadienyl-1H-indole (**1c**).* Yield 0.185 g, 93%, yellow oil. ^{1}H NMR (40 MHz, $CDCl_3$) δ 5.63 (d, J = 6.4 Hz, 2H), 6.64 (d, J = 3.2 Hz, 1H), 7.09−7.78 (m, 6H). Spectroscopic data are in accordance with those reported in the literature.[28]

*1-Propadienyl-1H-pyrrole (**1d**).* It was synthesized according to the literature procedure.[28] A total of 1.034 mL (14.925 mmol) of pyrrole was used. Yield 0.705 g, 45%, yellow oil. ^{1}H NMR (40 MHz, $CDCl_3$) δ 5.57 (d, J = 6.4 Hz, 2H), 6.18−6.45 (m, 2H), 6.80−7.45 (m, 3H).

*5-Methoxy-1-propadienyl-1H-indole (**1e**).* Yield 0.195 g, 82%, yellow crystallizing liquid. R_f = 0.41 (EtOAc−hexane 1−10). ^{1}H NMR (600 MHz, $CDCl_3$) δ 3.86 (s, 3H), 5.63 (d, J = 6.6 Hz, 2H), 6.53 (d, J = 3.2 Hz, 1H), 6.90 (dd, J = 2.5, 8.9 Hz, 1H), 7.09 (d, J = 2.5 Hz, 1H), 7.15−7.17 (m, 2H), 7.49 ppm (d, J = 8.9 Hz, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 55.8, 87.7, 97.7, 103.0, 103.6, 110.8, 112.2, 126.3, 129.9, 130.6, 154.6, 203.3. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{12}H_{12}NO$: 186.0913; found: 186.0913. IR (KBr): 3154−2854, 1723−1666, 1620−1580, 1478, 1448, 1247−1231, 1152−1131, 1029, 799, 716 cm^{-1}.

*5-Bromo-1-propadienyl-1H-indole (**1f**).* Yield 0.261 g, 87%, yellow crystallizing oil. R_f = 0.47 (EtOAc−hexane 1−10). ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.60 (d, J = 6.4 Hz, 2H), 6.50 (d, J = 3.3 Hz, 1H), 7.11 (t, J = 6.4 Hz, 1H), 7.16 (d, J = 3.3 Hz, 1H), 7.28 (dd, J = 2.0, 8.8 Hz, 1H), 7.40 (d, J = 8.8 Hz, 1H), 7.70 ppm (d, J = 2.0 Hz, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 87.9, 97.1, 103.3, 111.4, 113.4, 123.4, 124.9, 126.8, 130.8, 133.9, 202.9. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{11}H_9NBr$: 233.9913; found: 233.9919. IR (KBr): 3154−2854, 1723−1666, 1620−1580, 1478, 1448, 1247−1231, 1152−1131, 1029, 799, 716 cm^{-1}.

*6-Bromo-1-propadienyl-1H-indole (**1g**).* Yield 0.264 g, 88%, yellow crystallizing oil. R_f = 0.53 (EtOAc−hexane 1−10). ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.66 (d, J = 6.5 Hz, 2H), 6.58 (d, J = 3.1 Hz, 1H), 7.12 (t, J = 6.5 Hz, 1H), 7.18 (d, J = 3.1 Hz, 1H), 7.26 (dd, J = 1.5, 8.4 Hz, 1H), 7.47 (d, J = 8.4 Hz, 1H), 7.73 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 88.0, 97.1, 104.1, 113.1, 115.9, 122.2, 123.6, 126.4, 128.2, 136.1, 203.1. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{11}H_9NBr$: 233.9913; found: 233.9919. IR (KBr): 3143−3035, 2872−2836, 1958, 1875, 1602, 1514, 1463, 1320, 1229, 892, 802, 713 cm^{-1}.

*4-Cyano-1-propadienyl-1H-indole (**1h**).* Yield 0.208, 92%, white crystals. MP 71−72 °C. R_f = 0.15 (EtOAc−hexane 1−10). ^{1}H NMR (600 MHz, $CDCl_3$) ^{1}H NMR (600 MHz, $CDCl_3$) δ 5.67 (d, J = 6.5 Hz, 2H), 6.80 (d, J = 3.3 Hz, 1H), 7.18 (t, J = 6.5 Hz, 1H), 7.26 (d, J = 7.8 Hz, 1H), 7.36 (d, J = 3.3 Hz, 1H), 7.49 (d, J = 7.4 Hz, 1H), 7.78 ppm (d, J = 8.4 Hz, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 88.3, 96.9, 102.5, 103.4, 114.7, 118.3, 121.8, 125.4, 128.3, 130.5, 134.9, 202.9. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{12}H_9N_2$: 181.0760; found: 181.0760. IR (KBr): 3129−3088, 3047−2981, 2220, 1598, 1518, 1463, 1426, 1330, 1221, 1168, 1053, 899, 747 cm^{-1}.

*3-Methyl-1-propadienyl-1H-indole (**1i**).* Yield 0.206, 95%, yellow oil. R_f = 0.17 (EtOAc−hexane 1−10). ^{1}H NMR (600 MHz, DMSO-d_6) δ 2.23 (s, 3H), 5.69 (d, J = 6.1 Hz, 2H), 7.06−7.08 (m, 1H), 7.10 (c, 1H), 7.15−7.18 (m, 1H), 7.49 (d, J = 7.6 Hz, 1H), 7.64−7.66 (m, 1H), 7.67 ppm (d, J = 8.1 Hz, 1H). ^{13}C{^{1}H} NMR (150 MHz, DMSO-d_6) δ 9.3, 97.4, 110.3, 112.6, 118.9, 119.7, 122.2, 123.3, 129.3, 135.3, 202.4. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{12}H_{12}N$: 170.0964; found: 170.0964. IR (KBr): 3108−3014, 2974−2819, 1714−1651, 1613, 1462, 1390, 1338, 1227, 1189, 1126−1016, 954, 740 cm^{-1}.

Synthesis of the Various Starting Arylallenes 3a−h. *Propadienylbenzene (**3a**).* It was synthesized from 3-phenyl-1-propyne according to the literature procedure.[29] ^{1}H NMR (40 MHz, $CDCl_3$) δ 5.26 (d, J = 6.7 Hz, 2H), 6.18−6.49 (m, 1H), 7.31−7.58 (m, 5H). Spectroscopic data are in accordance with those reported in the literature.[29]

*2-Propadienylnaphthalene (**3b**).* It was synthesized according to the literature procedure.[30] ^{1}H NMR (40 MHz, $CDCl_3$) δ 5.22 (d, J = 6.7 Hz, 2H), 6.19−6.50 (m, 1H), 7.21−7.97 (m, 7H). Spectroscopic data are in accordance with those reported in the literature.[30]

*1-Methyl-4-propadienylbenzene (**3c**).* It was synthesized according to the literature procedure.[30] ^{1}H NMR (40 MHz, $CDCl_3$) δ 2.41 (s, 3H), 5.18 (d, J = 6.7 Hz, 2H), 6.07−6.39 (m, 1H), 7.23−7.32 (m, 4H). Spectroscopic data are in accordance with those reported in the literature.[30]

*1-Methoxy-4-propadienylbenzene (**3d**).* It was synthesized from 4-methoxystyrene according to the literature procedure.[30] ^{1}H NMR (40 MHz, $CDCl_3$) δ 3.77 (c, 3H), 5.09 (d, J = 6.7 Hz, 2H), 5.97−6.28 (m, 1H), 6.80 (d, J = 8.6 Hz, 2H), 7.23 (d, J = 8.6 Hz, 2H). Spectroscopic data are in accordance with those reported in the literature.[30]

*Buta-2,3-dien-1-ylbenzene (**3e**).* The compound was synthesized from benzylbromide and propargyl bromide according to the literature procedure.[31] ^{1}H NMR (100 MHz, $CDCl_3$) δ 3.27−3.57 (m, 2H), 4.61−4.91 (m, 2H), 5.05−5,69 (m, 1H), 7.31 (m, 5H). Spectroscopic data are in accordance with those reported in the literature.[31]

*(Propadienyloxy)benzene (**3f**).* It was synthesized from (prop-2-yn-1-yloxy)benzene according to the literature procedure.[32] ^{1}H NMR (40 MHz, $CDCl_3$) δ 5.47 (d, J = 6.0 Hz, 2H), 6.61−6.93 (m, 1H), 7.03−7.27 (m, 5H). Spectroscopic data are in accordance with those reported in the literature.[32]

*1-(Buta-2,3-dien-1-yloxy)-4-(propan-2-yl)benzene (**3g**).* To a mixture of 0.500 g of the alkyne (1 equiv, 2.873 mmol), 0.172 g of paraformaldehyde (2 equiv, 5.873 mmol), and 0.205 g of CuBr (0.5 equiv, 1.4365 mmol) in 14 mL of 1,4-dioxane, 0.728 mL of NH^iPr_2 (1.8 equiv, 5.1714 mmol) was added in one portion. The resulting bright-green-yellow mixture was then refluxed for 2 h and after cooling to rt was diluted with water and extracted with Et_2O. The organic phase was dried with sodium sulfate and evaporated under reduced pressure. Flash chromatography on silica gel (EtOAc−hexane 1−10) gave 0.303 g of (56%) of pure allene as a yellowish oil. R_f = 0.5 (hexane). ^{1}H NMR (600 MHz, $CDCl_3$) δ 1.25 (d, J = 6.9 Hz, 6H), 2.80 (spt, J = 6.9 Hz, 1H), 4.57 (m, 2H), 4.88 (dt, J = 2.5, 6.6 Hz, 2H), 5.42 (m, 1H), 6.88 (d, J = 8.5 Hz, 2H), 7.16 ppm (d, J = 8.5 Hz, 2H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 24.2 (2C), 33.2, 65.9, 76.4, 87.2, 114.6 (2C), 127.2 (2C), 141.3, 156.4, 209.4. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{13}H_{17}O$: 189.1274; found: 189.1274. IR (KBr): 2960, 2932−2869, 1958, 1610, 1511, 1237, 1178, 1018, 847−828 cm^{-1}.

*1-Propadienyl-4-(trifluoromethyl)benzene (**3h**).* It was synthesized from 4-trifluoromethylstyrene according to the literature procedure.[33] ^{1}H NMR (40 MHz, $CDCl_3$) δ 5.32 (d, J = 6.7 Hz, 2H), 6.17−6.48 (m, 1H), 7.36−7.82 (m, 4H). Spectroscopic data are in accordance with those reported in the literature.[33]

General Procedure for the Hydrosulfonylation of *N*-Indolylallenes. A mixture of an appropriate allene (**A**, g), 5 mol % of EY (**B**, g), 3 equiv of a sulfinate (**C**, g), and 2 equiv of K_2CO_3 (**D**, g) in a DMF−TFE mixture (1:0.3) as the solvent under a nitrogen atmosphere was subjected to green light irradiation (green LED) and stirred for the appropriate time (TLC). Upon the completion, the solvent was evaporated under reduced pressure and the residue was chromatographed on silica with EtOAc−hexane mixtures. After chromatography, the product was crystalized in a small amount of hexane with a few drops of EtOAc and then washed with the same mixture.

*2-Phenyl-1-[(1E)-2-(phenlsulfonyl)prop-1-en-1-yl]-1H-indole (**2a**).* **A**: 0.070 g (0.3043 mmol, 1 equiv) of **1a**, **B**: 0.0105 g (0.0152 mmol, 0.05 equiv) of EY, **C**: 0.162 g (0.9129 mmol, 3 equiv) of sodium *p*-toluene sulfinate, **D**: 0.084 g (0.6086 mmol, 2 equiv) of K_2CO_3, 2.542 mL of DMF, and 0.763 mL of TFE.

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Yield 0.074 g (63%), white crystals. MP = 152 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 1.69 (s, 3H), 2.51 (s, 3H), 6.77 (s, 1H), 7.18 (d, J = 8.2 Hz, 1H), 7.22−7.24 (m, 1H), 7.27−7.30 (m, 1H), 7.31−7.35 (m, 3H), 7.37−7.38 (m, 2H), 7.40 (d, J = 8.1 Hz, 2H), 7.65 (d, J = 7.8 Hz, 1H), 7.80 (d, J = 8.1 Hz, 2H), 8.03 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 13.2, 21.7, 105.6, 111.0, 121.0, 121.7, 123.1, 128.19, 128.22 (2C), 128.4 (2C), 128.7 (2C), 129.0, 130.0 (2), 132.0, 133.9, 134.6, 136.0, 137.1, 140.6, 144.7. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{24}H_{22}NO_2S$: 388.1366; found: 388.1360. IR (KBr): 3131−3031, 2948−2856, 1650, 1596, 1457, 1158, 1122, 1078, 741 cm^{-1}.

3-Methyl-1-[(1E)-2-(phenylsulfonyl)prop-1-en-1-yl]-1H-indole ***(2b)***. **A**: 0.096 g (0.5681 mmol, 1 equiv) of **1i**, **B**: 0.020 g (0.0284 mmol, 0.05 equiv) of EY, **C**: 0.303 g (1.7043 mmol, 3 equiv) of sodium *p*-toluene sulfinate, **D**: 0.157 g (1.1362 mmol, 2 equiv) of K_2CO_3, 4.745 mL of DMF, and 1.424 mL of TFE.

Yield 0.078 g (42%), white crystals. MP = 122 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.19 (s, 3H), 2.33 (s, 3H), 2.44 (s, 3H), 7.24 (s,1H), 7.26−7.30 (m, 1H), 7.33−7.39 (m, 3H), 7.56−7.60 (m, 2H), 7.82 (d, J = 7.9 Hz, 2H), 8.38 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 9.7, 12.7, 21.5, 110.0, 117.5, 118.7, 119.3, 122.0, 122.1, 123.8, 127.9 (2C), 129.4, 129.8 (2C), 131.1, 136.9, 137.3, 143.9. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{19}H_{20}NO_2S$: 326.1209; found: 326.1209. IR (KBr): 3162−3144, 3054, 2978−2926, 2883, 1642, 1596, 1460, 1400, 1351, 1306−1289, 1240, 1149, 1120, 1074, 737, 681, 586 cm^{-1}.

5-Bromo-1-[(1E)-2-(phenylsulfonyl)prop-1-en-1-yl]-1H-indole ***(2c)***. **A**: 0.137 g (0.5855 mmol, 1 equiv) of **1f**, **B**: 0.020 g (0.029 mmol, 0.05 equiv) of EY, **C**: 0.242 g (1.756 mmol, 3 equiv) of sodium *p*-toluene sulfinate, **D**: 0.162 g (1.1708 mmol, 2 equiv) of K_2CO_3, 4.890 mL of DMF, and 1.487 mL of TFE.

Yield 0.094 g (43%), white crystals. MP = 180 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.18 (s, 3H), 2.45 (s, 3H), 6.67 (d, J = 3.4 Hz, 1H), 7.36 (d, J = 8.1 Hz, 2H), 7.41 (d, J = 3.4 Hz, 1H), 7.45 (m, 2H), 7.76 (s, 1H), 7.82 (d, J = 8.1 Hz, 2H), 8.31 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 12.8, 21.6, 107.1, 111.6, 115.7, 122.6, 123.9, 126.2, 126.6, 128.0 (2C), 129.9 (2C), 130.2, 130.8, 135.6, 136.3, 144.4. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{18}H_{17}NO_2SBr$: 390.0158; found: 390.0163. IR (KBr): 3118−3029, 2985−2857, 1651, 1593, 1520, 1451, 1389, 1302, 1282, 1235, 1150, 1114, 1075, 962, 816, 802, 762, 747, 675, 560 cm^{-1}.

6-Bromo-1-[(1E)-2-(phenylsulfonyl)prop-1-en-1-yl]-1H-indole ***(2d)***. **A**: 0.137 g (0.5855 mmol, 1 equiv) of **1g**, **B**: 0.020 g (0.029 mmol, 0.05 equiv) of EY, **C**: 0.288 g (1.757 mmol, 3 equiv) of sodium benzene sulfinate, **D**: 0.162 g (1.1708 mmol, 2 equiv) of K_2CO_3, 4.890 mL of DMF, and 1.487 mL of TFE.

Yield 0.070 g (32%), white crystals. MP = 187 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.19 (s, 3H), 6.70 (d, J = 3.3 Hz, 1H), 7.37 (d, J = 8.4 Hz, 1H), 7.40 (d, J = 3.3 Hz, 1H), 7.49 (d, J = 8.4 Hz, 1H), 7.57−7.60 (m, 2H), 7.65 (t, J = 7.4 Hz, 1H), 7.75 (s, 1H), 7.96 (d, J = 7.8 Hz, 2H), 8.31 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 12.8, 107.8, 113.4, 117.4, 122.2, 122.5, 125.6, 125.8, 127.4, 128.0 (2C), 129.3 (2C), 131.0, 133.4, 137.7, 139.3. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{17}H_{15}NO_2SBr$: 376.0001; found: 376.0001. IR (KBr): 3175−2980, 1655, 1601, 1521, 1460, 1326, 1293, 1243, 1148, 886, 817, 757, 714, 961, 597, 569 cm^{-1}.

5-Methoxy-1-[(1E)-2-(phenylsulfonyl)prop-1-en-1-yl]-1H-indole ***(2e)***. **A**: 0.092 g (0.4973 mmol, 1 equiv) of **1e**, **B**: 0.017 g (0.025 mmol, 0.05 equiv) of EY, **C**: 0.266 g (1.492 mmol, 3 equiv) of sodium *p*-toluene sulfinate, **D**: 0.137 g (0.9946 mmol, 2 equiv) of K_2CO_3, 4.153 mL of DMF, and 1.246 mL of TFE.

Yield 0.077 g (45%), white crystals. MP = 129 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.18 (s, 3H), 2.45 (s, 3H), 3.88 (s, 3H), 6.66 (d, J = 3.4 Hz, 1H), 7.00 (dd, J = 2.4, 8.9 Hz, 2H), 7.09 (d, J = 2.4 Hz, 1H), 7.35 (d, J = 8.1 Hz, 2H), 7.41 (d, J = 3.4, 1H), 7.49 (d, J = 8.9 Hz, 1H), 7.82 (d, J = 8.1 Hz, 2H), 8.34 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 12.7, 21.6, 55.8, 103.4, 107.7, 110.9, 113.2, 120.2, 125.5, 128.0 (2C), 129.3, 129.9 (2C), 131.3, 131.9, 136.7, 141.1, 156.0. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{19}H_{20}NO_3S$: 342.1158; found: 342.1158. IR (KBr): 3118−3029, 2996−2845, 1644, 1617, 1596, 1476, 1453, 1407, 1296, 1262, 1148, 1076, 962, 800, 751, 718, 649, 582 cm^{-1}.

1-{(1E)-2-[(4-Methylphenyl)sulfonyl]prop-1-en-1-yl}-1H-indole-4-carbonitrile ***(2f)***. **A**: 0.090 g (0.500 mmol, 1 equiv) of **1h**, **B**: 0.017 g (0.025 mmol, 0.05 equiv) of EY, **C**: 0.267 g (1.500 mmol, 3 equiv) of sodium *p*-toluene sulfinate, **D**: 0.138 g (1.00 mmol, 2 equiv) of K_2CO_3, 4.176 mL of DMF, and 1.253 mL of TFE.

Yield 0.069 g (41%), white crystals. MP = 181−182 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.19 (s, 3H), 2.46 (s, 3H), 6.94 (d, J = 3.4 Hz, 1H), 7.37 (d, J = 8.1 Hz, 2H), 7.40−7.43 (m, 1H), 7.57 (d, J = 3.4 Hz, 1H), 7.60 (d, J = 7.5 Hz, 1H), 7.80 (d, J = 8.4, 1H), 7.83 (d, J = 8.1 Hz, 2H), 8.34 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 12.8, 21.6, 104.1, 105.8, 114.8, 117.7, 123.4, 125.0, 127.2, 127.8, 128.1 (2C), 130.0 (2C), 130.1, 130.2, 135.9, 136.5, 144.7. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{19}H_{17}N_2O_2S$: 337.1005; found: 337.1002. IR (KBr): 3173, 3123−3034, 2988−2924, 2871, 2228, 1649, 1595, 1521, 1440, 1408, 1294, 1153, 1076, 749, 660, 598 cm^{-1}.

6-Bromo-1-[(1E)-2-(thiophen-2-ylsulfonyl)prop-1-en-1-yl]-1H-indole ***(2g)***. **A**: 0.137 g (0.5855 mmol, 1 equiv) of **1g**, **B**: 0.020 g (0.029 mmol, 0.05 equiv) of EY, **C**: 0.299 g (1.757 mmol, 3 equiv) of sodium thiophene-2-sulphinate, **D**: 0.162 g (1.1708 mmol, 2 equiv) of K_2CO_3, 4.890 mL of DMF, and 1.487 mL of TFE.

Yield 0.065 g (29%), white crystals. MP = 191−192 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.30 (s, 3H), 6.71 (d, J = 3.4 Hz, 1H), 7.16−7.17 (m, 1H), 7.35 (dd, J = 1.3, 8.5 Hz, 1H), 7.42 (d, J = 3.4 Hz, 1H), 7.48 (d, J = 8.5 Hz, 1H), 7.71−7.75 (m, 3H), 8.29 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 12.8, 107.9, 113.4, 117.4, 122.5, 122.7, 125.5, 125.8, 127.5, 127.9, 130.8, 133.9, 134.0, 137.7, 140.8. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{15}H_{13}NO_2S_2Br$: 381.9566; found: 381.9566. IR (KBr): 3165, 3113−3089, 2978−2857, 1650, 1601, 1521, 1461, 1437, 1401, 1326, 1296, 1242, 1148, 1109, 1013, 887, 816, 756, 718, 675, 600, 572 cm^{-1}.

1-{(1E)-2-[(4-Methylphenyl)sulfonyl]prop-1-en-1-yl}-1H-indole ***(2h)***. **A**: 0.072 g (0.5548 mmol, 1 equiv) of **1c**, **B**: 0.019 g (0.0277 mmol, 0.05 equiv) of EY, **C**: 0.296 g (1.6644 mmol, 3 equiv) of sodium *p*-toluene sulfinate, **D**: 0.153 g (1.110 mmol, 2 equiv) of K_2CO_3, 4.634 mL of DMF, and 1.390 mL of TFE.

Yield 0.072 g (42%), white crystals. MP = 174 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.20 (s, 3H), 2.45 (s, 3H), 6.74 (d, J = 3.3 Hz, 1H), 7.26−7.29 (m, 1H), 7.34−7.39 (m, 3H), 7.43 (d, J = 3.3 Hz, 1H), 7.60 (d, J = 8.2 Hz, 2H), 7.64 (d, J = 7.8 Hz, 1H), 7.83 (d, J = 8.2 Hz, 2H), 8.41 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 12.8, 21.6, 107.8, 110.1, 121.1, 121.3, 122.5, 123.8, 125.0, 128.0 (2C), 128.6, 129.9 (2C), 131.2, 136.6, 137.0, 144.2. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{18}H_{18}NO_2S$: 312.1053; found: 312.1053. IR (KBr): 3165, 3114−3089, 2978−2857, 1650, 1601, 1521, 1461−1401, 1242, 1148, 1351, 1013, 887, 816, 756, 718, 675, 600, 572 cm^{-1}.

1-[(1E)-2-(Phenylsulfonyl)prop-1-en-1-yl]-1H-indole ***(2i)***. **A**: 0.090 g (0.5807 mmol, 1 equiv) of **1c**, **B**: 0.020 g (0.0290 mmol, 0.05 equiv) of EY, **C**: 0.286 g (1.7421 mmol, 3 equiv) of sodium benzene sulfinate, **D**: 0.160 g (1.1614 mmol, 2 equiv) of K_2CO_3, 4.850 mL of DMF, and 1.455 mL of TFE.

Yield 0.051 g (30%), white crystals. MP = 133 °C. ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.21 (s, 3H), 6.75 (d, J = 3.3 Hz, 1H), 7.27−7.29 (m, 1H), 7.38 (t, J = 7.7 Hz, 1H), 7.44 (d, J = 3.3 Hz, 1H), 7.56−7.58 (m, 2H), 7.60−7.65 (m, 3H), 7.96 (d, J = 7.8 Hz, 2H), 8.44 ppm (s, 1H). ^{13}C{^{1}H} NMR (150 MHz, $CDCl_3$) δ 12.7, 108.0, 110.1, 120.6, 121.3, 122.5, 123.8, 124.9, 127.9 (2C), 128.6, 129.12 (2C), 131.6, 133.2, 137.0, 139.6. HRMS (ESI): m/z [M + H]$^+$ calcd for $C_{17}H_{16}NO_2S$: 298.0896; found: 298.0896. IR (KBr): 3167, 3116−3050, 3017, 2976, 2917−2861, 1650, 1582, 1530, 1463−1446, 1409, 1332, 1302, 1242, 1149−1116, 1077, 964, 743, 687, 619, 567 cm^{-1}.

1-[(1E)-2-(Methylsulfonyl)prop-1-en-1-yl]-1H-indole ***(2j)***. **A**: 0.090 g (0.5807 mmol, 1 equiv) of **1a**, **B**: 0.020 g (0.0290 mmol, 0.05 equiv) of EY, **C**: 0.177 g (1.7421 mmol, 3 equiv) of sodium methanesulfinate, **D**: 0.160 g (1.1614 mmol, 2 equiv) of K_2CO_3, 4.850 mL of DMF, and 1.455 mL of TFE.

Yield 0.053 g (39%), yellowish oil. R_f = 0.33 (EtOAc−hexane 1−2). ^{1}H NMR (600 MHz, $CDCl_3$) δ 2.42 (s, 3H), 3.02 (s, 3H), 6.78 (d, J = 3.4 Hz, 2H), 7.26−7.28 (m, 1H), 7.33−7.35 (m, 1H), 7.50 (d,

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J = 3.4 Hz, 1H), 7.53 (d, J = 8.2 Hz, 1H), 7.64 (d, J = 7.7 Hz, 1H), 8.23 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 12.9, 41.3, 108.2, 100.0, 121.3, 121.4, 122.5, 123.8, 124.8, 128.5, 132.0, 136.9. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{12}H_{14}NO_2S$: 236.0740; found: 236.0740. IR (KBr): 3179−2989, 2927, 1654, 1460, 1327, 1295, 1241, 1145, 1102, 966, 774, 743, 561 cm^{-1}.

*5-Methoxy-1-[(1E)-2-(thiophen-2-ylsulfonyl)prop-1-en-1-yl]-1H-indole (**2k**).* **A**: 0.100 g (0.5405 mmol, 1 equiv) of **1e**, **B**: 0.019 g (0.027 mmol, 0.05 equiv) of EY, **C**: 0.275 g (1.620 mmol, 3 equiv) of sodium thiophene-2-sulphinate, **D**: 0.149 g (0.1081 mmol, 2 equiv) of K_2CO_3, 4.513 mL of DMF, and 1.354 mL of TFE.

Yield 0.045 g (25%), white crystals. MP = 158−160 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 2.30 (s, 3H), 3.87 (s, 3H), 6.68 (d, J = 3.4 Hz, 1H), 6.99 (dd, J = 2.4, 8.9 Hz, 2H), 7.09 (d, J = 2.4 Hz, 1H), 7.14−7.15 (m, 1H), 7.44 (d, J = 3.4 Hz, 1H), 7.47 (d, J = 8.9, 1H), 7.69 (d, J = 4.9 Hz, 1H), 7.72 (d, J = 3.8 Hz, 1H), 8.35 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 12.8, 29.7, 103.5, 108.1, 110.8, 113.2, 120.3, 125.4, 127.7, 129.3, 131.5, 131.9, 133.5, 133.6, 141.3, 156.1. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{16}H_{16}NO_3S_2$: 334.0566; found: 334.0572. IR (KBr): 3118−3051, 3007−2830, 1651, 1614, 1476, 1404, 1269, 1264, 1143−1132, 751−717, 596, 574 cm^{-1}.

*2-(4-Methylphenyl)-1-[(1E)-2-(phenylsulfonyl)prop-1-en-1-yl]-1H-indole (Major) and 2-(4-Methylphenyl)-1-{2-[(4-methylphenyl)-sulfonyl]prop-2-en-1-yl}-1H-indole (Minor) (**2l + 2l′**).* **A**: 0.087 g (0.3551 mmol, 1 equiv) of **1b**, **B**: 0.012 g (0.0178 mmol, 0.05 equiv) of EY, **C**: 0.190 g (1.0653 mmol, 3 equiv) of sodium p-toluene sulfinate, **D**: 0.098 g (0.7102 mmol, 2 equiv) of K_2CO_3, 2.966 mL of DMF, and 0.890 mL of TFE.

Yield 0.075 g (53%), white crystals. MP = 168 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 1.69 (s, 3H), 2.39−2.41 (m, 4.2H), 2.52 (s, 4.2H), 4.91 (s, 0.82H), 5.21 (s, 0.4H), 6.39 (s, 0.4H), 6.57 (s, 0.4H), 6.74 (s, 1H), 6.88 (d, J = 6.4 Hz, 0.4H), 7.07 (d, J = 7.8 Hz, 0.8H), 7.12−7.17 (m, 4H), 7.19 (d, J = 8.2 Hz, 1H), 7.22−7.24 (m, 1H), 7.26−7.30 (m, 3.8H), 7.37 (d, J = 8.1 Hz, 0.8H), 7.41 (d, J = 8.2 Hz, 2H), 7.62 (dd, J = 2.4, 5.6 Hz, 0.4H), 7.65 (d, J = 7.6 Hz, 1H), 7.77 (d, J = 8.1 Hz, 0.4H), 7.82 (d, J = 8.2 Hz, 2H), 8.04 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.2, 21.2, 21.61, 21.64, 43.1, 102.6, 105.0, 109.5, 110.9, 120.5, 120.6, 120.8, 121.5, 122.1, 122.9, 124.1, 128.15, 128.18 (2C), 128.2 (2C), 128.3 (2C), 128.7, 129.0 (2C), 129.2 (2C), 129.3 (2C), 129.5, 129.9 (2C), 130.0 (2C), 133.8, 133.4, 134.4, 135.4, 136.0, 137.0, 137.2, 138.0, 140.7, 141.1, 144.5, 144.9, 147.0. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{25}H_{23}NO_2S$: 402.1522; found: 402.1522. IR (KBr): 3080−3030, 2992−2865, 1651, 1594, 1503, 1454, 1364, 1314, 1161, 1078, 964, 882, 797, 741, 655 cm^{-1}.

*1-[(1E)-2-(Phenylsulfonyl)prop-1-en-1-yl]-1H-pyrrole (**2m**).* **A**: 0.500 g (4.760 mmol, 1 equiv) of **1d**, **B**: 0.165 g (0.238 mmol, 0.05 equiv) of EY, **C**: 2.342 g (14.280 mmol, 3 equiv) of sodium benzene sulfinate, **D**: 1.314 g (9.520 mmol, 2 equiv) of K_2CO_3, 20 mL of DMF, and 6 mL of TFE.

Yield 0.410 g (35%), white crystals. MP = 119 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 2.14 (s, 3H), 6.34−6.35 (m, 2H), 7.00−7.01 (m, 2H), 7.54−7.56 (m, 2H), 7.62 (d, J = 7.5 Hz, 2H), 7.91 (d, J = 7.8 Hz, 2H), 8.03 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 12.5, 112.2 (2C), 120.9, 121.9 (2C), 127.8 (2C), 129.2 (2C), 133.2, 135.0, 139.4 ppm. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{13}H_{14}NO_2S$: 248.0739; found: 248.0737. IR (KBr): 3173, 3136, 3118, 3086, 3018, 980, 2936, 2872, 1657, 1569, 1542, 1489, 1449, 1335, 1298−1282, 1147, 1117, 1074, 1049, 943, 751, 692, 579 cm^{-1}.

General Procedure for the Hydrosulfonylation of Arylallenes. A mixture of an appropriate allene (**A** g), 5 mol % EY (**B**, g), 3 equiv of a sulfinate (**C**, g) in a DMF−TFE mixture (1:0.3) as the solvent under a nitrogen atmosphere was subjected to green light irradiation (green LED) and stirred for the appropriate time (TLC). Upon the completion, the solvent was evaporated under reduced pressure and the residue was chromatographed on silica with EtOAc−hexane mixtures in different proportions.

*1-Methyl-4-{[(1E)-1-phenylprop-1-en-2-yl]sulfonyl}benzene (**4a**).* **A**: 0.088 g (0.759 mmol, 1 equiv) of **3a**, B: 0.026 g (0.0380 mmol, 0.05 equiv) of EY, **C**: 0.405 g (2.276 mmol, 3 equiv) of sodium p-toluene sulfinate, 6.34 mL of DMF, and 1.9 mL of TFE.

Yield 0.141 g (68%), white crystals. ^1H NMR (600 MHz, $CDCl_3$) δ 2.12 (s, 3H), 2.45 (s, 3H), 7.34−7.43 (m, 7H), 7.81−7.82 ppm (m, 3H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.1, 21.6, 128.2 (2C), 128.6 (2C), 129.2, 129.6 (2C), 129.8 (2C), 133.8, 136.0, 137.0, 137.5, 144.2. Spectroscopic data are in accordance with those reported in the literature.[34]

*{[(1E)-1-Phenylprop-1-en-2-yl]sulfonyl}benzene (**4b**).* **A**: 0.09 g (0.776 mmol, 1 equiv) of **3a**, **B**: 0.027 g (0.0388 mmol, 0.05 equiv) of EY, **C**: 0.394 g (2.328 mmol, 3 equiv) of sodium benzene sulfinate, 6.48 mL of DMF, and 1.94 mL of TFE.

Yield 0.156 g (78%), white crystals. ^1H NMR (600 MHz, $CDCl_3$) δ 2.13 (s, 3H), 7.38−7.43 (m, 5H), 7.55−7.58 (m, 2H), 7.63 (t, J = 7.3 Hz, 1H), 7.84 (s, 1H), 7.94 ppm (d, J = 7.7 Hz, 2H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.2, 128.1 (2C), 128.7 (2C), 129.2 (2C), 129.3, 129.6 (2C), 133.3, 133.7, 137.2, 137.5, 139.0. Spectroscopic data are in accordance with those reported in the literature.[35]

*1-Bromo-4-{[(1E)-1-phenylprop-1-en-2-yl]sulfonyl}benzene (**4c**).* **A**: 0.07 g (0.6034 mmol, 1 equiv) of **3a**, B: 0.021 g (0.03017 mmol, 0.05 equiv) of EY, **C**: 0.440 g (1.810 mmol, 3 equiv) of sodium 4-bromobenzene sulfinate, 5.039 mL of DMF, and 1.311 mL of TFE.

Yield 0.150 g (74%), white crystals. MP = 122 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 2.13 (s, 3H), 7.38−7.44 (m, 5H), 7.70 (d, J = 8.5 Hz, 2H), 7.80 (d, J = 8.5 Hz, 2H), 7.83 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.2, 128.6, 128.7 (2C), 129.5, 129.6 (2C), 129.7 (2C), 132.5 (2C), 133.5, 136.8, 138.0, 138.2. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{15}H_{14}O_2SBr$: 336.9892; found: 336.9895. IR (KBr): 3089, 3071, 3030, 2965−2922, 1920, 1570, 1466, 1447, 1386, 1309, 1275, 1151, 1103, 1065, 823, 749, 691, 629, 577 cm^{-1}.

*(1E)-1-Phenylprop-1-en-2-yl thiophen-2-yl sulfone (**4e**).* **A**: 0.062 g (0.539 mmol, 1 equiv) of **3a**, **B**: 0.019 g (0.02695 mmol, 0.05 equiv) of EY, **C**: 0.275 g (1.6164 mmol, 3 equiv) of sodium tiophene-2-sulfinate, 4.5 mL of DMF, and 1.35 mL of TFE.

Yield 0.069 g (48%), white crystals. MP = 98 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 2.44 (s, 3H), 7.15−7.16 (m, 1H), 7.37−7.44 (m, 5H), 7.71−7.73 (m, 2H), 7.82 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.2, 127.8, 128.7 (2C), 129.4, 129.7 (2C), 133.7, 133.9, 134.1, 137.3, 137.8, 140.5. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{13}H_{13}O_2S_2$: 265.0351; found: 265.0351. IR (KBr): 3109−3036, 2963−2923, 1630, 1504, 1401, 1345, 1307, 1228, 1148, 1106, 1090, 1014, 770, 731, 700, 623, 570 cm^{-1}.

*[(1E)-2-(Methylsulfonyl)prop-1-en-1-yl]benzene (**4f**).* **A**: 0.07 g (0.6034 mmol, 1 equiv) of **3a**, B: 0.021 g (0.03017 mmol, 0.05 equiv) of EY, **C**: 0.185 g (1.81 mmol, 3 equiv) of sodium methanesulfinate, 5.039 mL of DMF, and 1.511 mL of TFE.

Yield 0.097 g (82%), yellowish oil. ^1H NMR (600 MHz, $CDCl_3$) δ 2.33 (s, 3H), 2.97 (s, 3H), 7.38−7.45 (m, 5H), 7.64 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.4, 40.3, 128.7 (2C), 129.4, 129.5 (2C), 133.5, 136.7, 137.9. Spectroscopic data are in accordance with those reported in the literature.[36]

*1-Methyl-4-{[(1E)-1-(4-methylphenyl)prop-1-en-2-yl]sulfonyl}-benzene (**4g**).* **A**: 0.100 g (0.769 mmol, 1 equiv) of **3c**, B: 0.027 g (0.03845 mmol, 0.05 equiv) of EY, **C**: 0.411 g (2.307 mmol, 3 equiv) of sodium p-toluene sulfinate, 6.422 mL of DMF, and 1.927 mL of TFE.

Yield 0.143 g (65%), white crystals. MP = 116 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 2.11 (s, 3H), 2.38 (s, 3H), 2.44 (s, 3H), 7.22 (d, J = 7.8 Hz, 2H), 7.31 (d, J = 7.8 Hz, 2H), 7.34 (d, J = 8.0 Hz, 2H), 7.78 (s, 1H), 7.80 ppm (d, J = 8.0 Hz, 2H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.2, 21.3, 21.5, 128.1 (2C), 129.3 (2C), 129.6 (2C), 129.7 (2C), 131.0, 136.2, 136.3, 137.0, 139.5, 144.1. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{17}H_{19}O_2S$: 287.1100; found: 287.1103. IR (KBr): 3087, 3030, 2984, 2923, 2869, 1916, 1633, 1598, 1511, 1445, 1297, 1148, 1104, 1072, 908, 811, 741, 654, 584 cm^{-1}.

*(1E)-1-(Naphthalen-2-yl)prop-1-en-2-yl phenyl sulfone (**4h**).* **A**: 0.075 g (0.4518 mmol, 1 equiv) of **3b**, B: 0.016 g (0.02259 mmol, 0.05 equiv) of EY, **C**: 0.222 g (1.3554 mmol, 3 equiv) of sodium benzene sulfinate, 3.77 mL of DMF, and 1.13 mL of TFE.

Yield 0.111 g (76%), white crystals. MP = 126 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 2.21 (s, 3H), 2.46 (s, 3H), 7.37 (d, J = 8.0 Hz, 2H), 7.50 (d, J = 8.7 Hz, 1H), 7.53−7.55 (m, 2H), 7.85−7.88 (m, 6H),

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7.97 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.3, 21.6, 126.4, 126.7, 127.2, 127.7, 128.27 (2C), 128.34 (2C), 129.7, 129.9 (2C), 131.3, 133.0, 133.3, 136.1, 137.1, 137.7 144.3. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{20}H_{19}O_2S$: 323.1100; found: 323.1102. IR (KBr): 3082, 3053, 3027, 2971−2924, 1622, 1594, 1493, 1449, 1301, 1143, 1078, 817, 740, 655, 589 cm^{-1}.

*1-Methyl-4-{[(1E)-1-phenoxyprop-1-en-2-yl]sulfonyl}benzene (**4i**).* **A:** 0.076 g (0.5757 mmol, 1 equiv) of **3f**, B: 0.020 g (0.02879 mmol, 0.05 equiv) of EY, **C:** 0.307 g (1.7271 mmol, 3 equiv) of sodium *p*-toluene sulfinate, 4.634 mL of DMF, and 1.390 mL of TFE.

Yield 0.028 g (17%), white crystallizing oil. MP = 88−89 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 1.92 (s, 3H), 2.45 (s, 3H), 7.09 (d, J = 8.0 Hz, 2H), 7.20 (t, J = 7.5 Hz, 1H), 7.35 (d, J = 7.9 Hz, 2H), 7.38−7.40 (m, 2H), 7.79 (d, J = 8.0 Hz, 2H), 7.82 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 9.4, 21.6, 115.3, 117.2, 120.2, 124.8, 127.8 (2C), 129.8 (2C), 130.0 (2C), 137.0, 144.0, 151.2, 156.5. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{16}H_{17}O_3S$: 289.0893; found: 289.0890. IR (KBr): 3092−3015, 2961−2862, 1658, 1593, 1490, 1305, 1225, 1141, 1077, 765, 665, 603 cm^{-1}.

*1-Methoxy-4-{[(1E)-1-(phenyl)prop-1-en-2-yl]sulfonyl}benzene (**4l**).* **A**: 0.082 g (0.5616 mmol, 1 equiv) of **3d**, **B**: 0.019 g (0.02808 mmol, 0.05 equiv) of EY, **C**: 0.276 g (2.307 mmol, 3 equiv) of sodium benzene sulfinate, 4.690 mL of DMF, and 1.407 mL of TFE.

Yield 0.121 g (74%), colorless oil. ^1H NMR (600 MHz, $CDCl_3$) δ 2.17 (s, 3H), 3.82 (s, 3H), 6.93 (d, J = 8.6 Hz, 2H), 7.38 (t, J = 8.6 Hz, 2H), 7.52−7.54 (m, 2H), 7.60 (t, J = 7.3 Hz, 1H), 7.76 (s, 1H), 7.91 ppm (d, J = 7.7 Hz, 2H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.2, 55.2, 114.1 (2C), 126.1, 127.9 (2C), 129.0 (2C), 131.4 (2C), 133.0, 134.3, 137.1, 139.3, 160.4. Spectroscopic data are in accordance with those reported in the literature.[37]

*1-[(1E)-2-(Phenylsulfonyl)prop-1-en-1-yl]-4-(trifluoromethyl)-benzene (**4m**).* **A**: 0.067 g (0.3641 mmol, 1 equiv) of **3h**, B: 0.013 g (0.0182 mmol, 0.05 equiv) of EY, **C**: 0.179 g (1.0923 mmol, 3 equiv) of sodium benzene sulfinate, 3.050 mL of DMF, and 0.900 mL of TFE.

Yield 0.082 g (69%), white crystals. MP = 102−103 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 2.11 (s, 3H), 7.50 (d, J = 8.1 Hz, 2H), 7.57−7.60 (m, 2H), 7.65−7.68 (m, 3H), 7.85 (s, 1H), 7.95 ppm (d, J = 8.1 Hz, 2H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.1, 121.0 (q, J_{CF} = 273.1 Hz), 125.6 (q, J_{CF} = 4.3 Hz, 2C), 128.3 (2C), 129.3 (2C), 129.7 (2C), 130.6 (q, J_{CF} = 33.2 Hz), 133.6, 135.7, 137.3, 138.5, 139.9. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{16}H_{14}F_3O_2S$: 327.0661; found: 327.0661. IR (KBr): 3108−3062, 2982−2928, 1732, 1615, 1447, 1412, 1328, 1299, 1154−1115, 1069, 968, 909, 856, 828, 749, 646, 579 cm^{-1}.

*1-Bromo-4-({(1E)-1-[4-(trifluoromethyl)phenyl]prop-1-en-2-yl}-sulfonyl)benzene (**4n**).* **A**: 0.067 g (0.3641 mmol, 1 equiv) of **3h**, **B**: 0.013 g (0.0182 mmol, 0.05 equiv) of EY, **C:** 0.265 g (1.0923 mmol, 3 equiv) of sodium 4-bromobenzene sulfinate, 3.050 mL of DMF, and 0.900 mL of TFE.

Yield 0.084 g (59%), white crystals. MP = 121 °C. ^1H NMR (600 MHz, $CDCl_3$) δ 2.09 (s, 3H), 7.47 (d, J = 8.1 Hz, 2H), 7.66 (t, J = 8.1 Hz, 2H), 7.71 (d, J = 8.6 Hz, 2H), 7.78 (d, J = 8.6 Hz, 2H), 7.82 ppm (s, 1H). $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$) δ 13.2, 121.0 (q, J_{CF} = 271.7 Hz), 125.7 (q, J_{CF} = 4.3 Hz, 2C), 129.0, 129.7 (2C), 129.8 (2C), 130.8 (q, J_{CF} = 31.8 Hz), 132.7 (2C), 136.2, 137.1, 137.6, 139.5. HRMS (ESI): m/z $[M + H]^+$ calcd for $C_{16}H_{13}F_3BrO_2S$: 404.9766; found: 404.9766. IR (KBr): 3143−3036, 2982−2802, 1732, 1628, 1574, 1325−1266, 1152, 1119, 1071, 1012, 929, 820, 758, 652, 587 cm^{-1}.

Scale-Up Experiment Procedure. *{[(1E)-1-Phenylprop-1-en-2-yl]sulfonyl}benzene (**4b**).* A mixture of 0.232 g of phenylallene **3a** (2 mmol, 1 equiv), 5 mol % EY (0.070 g, 0.1 mmol, 0.05 equiv), 3 equiv of sodium benzene sulfinate (0.984 g, 6 mmol) in 16.660 mL of DMF and 5.0 mL of TFE mixture (1:0.3) as the solvent under a nitrogen atmosphere was subjected to green light irradiation (green LED) and stirred for 5 h (TLC). Upon the completion, the solvent was evaporated under reduced pressure and the residue was chromatographed on silica with EtOAc−hexane mixture (1:5). Yield 0.418 g (81%), white crystals.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.9b02960.

Control experiments, X-ray crystallography data, and ^1H and ^{13}C spectra (PDF)

Crystallographic data (CIF)

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Notes

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